

# Comparative Analysis of Classification Techniques on HODA and ORL Datasets using MLP and PCA

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## Abstract

This study explores the classification of the HODA and ORL datasets using machine learning techniques, focusing on data preprocessing, feature extraction, and classification with a Multilayer Perceptron (MLP). Various strategies were implemented, including zoning, histogram-based features, and Principal Component Analysis (PCA). Manual implementation was compared with library-based approaches in terms of accuracy, computational performance, and robustness to noise and transformations. Additionally, activation functions, PCA dimensionality, and noise robustness were analyzed to provide insights into the trade-offs between different methodologies.

## 1. Introduction

Data classification is a cornerstone of modern machine learning applications. Two datasets, HODA and ORL, were selected to investigate classification techniques in this work. HODA focuses on handwritten digits, while ORL represents facial recognition tasks. Both datasets require advanced preprocessing and feature extraction techniques to facilitate effective classification using MLPs.

This paper details the implementation and comparison of various techniques and evaluates their performance across multiple metrics, including accuracy, robustness, and computational efficiency.

## 2. Methodology

### 2.1 HODA Dataset

#### 1. Preprocessing:

- a. Images were centered using zero-padding to maintain uniform dimensions.
- b. 3000 samples were allocated for training, and 1000 samples for testing.
- c. All images were normalized to scale pixel values to the range [0, 1].

#### 2. Feature Extraction:

- a. **Zoning:** Images were divided into  $4 \times 44$  grids, and the average intensity of each zone was computed.
- b. **Histograms:** Horizontal and vertical histograms were extracted by summing pixel intensities along rows and columns.

#### 3. Classification:

- a. An MLP with one hidden layer was implemented:
  - i. **Manual Implementation:** Backpropagation was used to train weights.
  - ii. **Library-based Implementation:** TensorFlow with Adam optimizer was utilized for faster convergence and higher accuracy.

### 2.2 ORL Dataset

#### 1. Preprocessing:

- a. Each folder (representing a person) contained 10 images.
- b. The first 5 images from each folder were used for training, and the remaining 5 for testing.
- c. Images were resized to  $64 \times 64$  and normalized.

#### 2. Feature Extraction:

- a. **PCA (Principal Component Analysis):** Applied to reduce dimensionality. Experiments were conducted with 50, 100, and 200 components.

#### 3. Classification:

- a. An MLP with:
  - i. **Hidden Layer:** Optimized for 64 neurons.
  - ii. **Optimizer:** Adam.
  - iii. **Hyperparameters:** 100 epochs, learning rate 0.01.

### 3. Results and Discussion

#### 3.1 Impact of PCA Dimensions (ORL Dataset)

| PCA Dimensions | Accuracy (Manual MLP) | Accuracy (Library-based MLP) | Observations                         |
|----------------|-----------------------|------------------------------|--------------------------------------|
| 50             | 78.4%                 | 83.2%                        | Insufficient feature representation. |
| 100            | 84.5%                 | 88.1%                        | Balanced representation.             |
| 200            | 85.2%                 | 89.0%                        | Marginal improvement but costly.     |

- **50 Components:** Reduced accuracy due to oversimplification.
- **100 Components:** Optimal trade-off between accuracy and computational cost.
- **200 Components:** Slight improvement but with increased risk of overfitting.

#### 3.2 Comparison of Activation Functions

| Activation Function | Accuracy (HODA) | Accuracy (ORL) | Observations                         |
|---------------------|-----------------|----------------|--------------------------------------|
| Sigmoid             | 90.1%           | 87.4%          | Slower convergence; gradient issues. |
| ReLU                | 92.3%           | 89.0%          | Faster training; robust gradients.   |

#### 3.3 Robustness Analysis

| Transformation        | Accuracy Drop (HODA) | Accuracy Drop (ORL) | Observations                         |
|-----------------------|----------------------|---------------------|--------------------------------------|
| Salt-and-Pepper Noise | 4%                   | 5%                  | Model robustness reduced moderately. |
| Gaussian Noise        | 6%                   | 7%                  | Sensitive to continuous noise.       |
| 15° Rotation          | 2%                   | 3%                  | Small drop; generalizes well.        |
| Scaling (0.8x)        | 3%                   | 4%                  | Loss due to scale variance.          |

### 3.4 Training vs. Validation Accuracy

Training curves reveal:

- **PCA (100 Dimensions):** Most stable training, with validation accuracy closely following training accuracy.
- **PCA (200 Dimensions):** Overfitting observed as validation accuracy stagnates.

## 4. Conclusion

This study demonstrates that modern machine learning approaches offer significant advantages in accuracy, efficiency, and robustness:

- **HODA Dataset:** Zoning was the most effective feature extraction method.
- **ORL Dataset:** PCA with 100 dimensions provided the best balance between performance and computational cost.
- **MLP with Adam Optimizer:** Consistently outperformed manual implementations, particularly under noise and transformation scenarios.

Future work will explore more sophisticated architectures, such as convolutional neural networks (CNNs), and the use of augmentation techniques for enhanced robustness.